DAY 27 | POST-READ

Protocols and Advanced Agents: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
MCP	Open protocol connecting AI applications to tools and data.
MCP roles	Host (the app), client (connector), server (exposes capabilities).
MCP primitives	Tools (actions), resources (data), prompts (templates). Transports: stdio, HTTP.
FastMCP	@mcp.tool(), @mcp.resource(uri), mcp.run(); type hints become schemas.
A2A	Agent-to-agent protocol: agent cards, tasks, messages, artifacts.
MCP vs A2A	Agent to tools vs agent to agent. They complement each other.
Handoff payload	Summary, facts, steps tried, reason, suggested next step, urgency.
Data analysis agent	Generate code, check the AST, run in a sandbox, show code and result.

Part B: Going beyond today

1. MCP SDK versions

The Python MCP SDK has evolved quickly; version 2 renamed FastMCP and changed some APIs. Today's notebook pins mcp below version 2 for stability. Check the official SDK documentation before starting a new project.

2. Securing MCP servers

Treat every MCP server like an API: authenticate remote servers, give tools least privilege, validate inputs, log calls, and review third-party servers before installing them. A malicious server can feed instructions to your model through tool results.

3. Remote MCP over HTTP

```
mcp.run(transport="streamable-http")   # serve on HTTP instead of stdio (FastMCP v1)
```

4. Managed code execution

Several providers offer hosted code interpreter tools that run model-written code in isolated containers. For customer work, prefer these or your own container sandbox over in-process

execution.

5. Discovering A2A agents

Agent cards make capabilities discoverable, like an API catalogue. Enterprises can keep a registry of approved internal agents and their skills.

Part C: Check yourself

- What problem does MCP solve?
- What is the difference between an MCP tool and a resource?
- What does an A2A agent card contain?
- List the six parts of a good handoff payload.
- Why show the generated code alongside a data agent's answer?

Part D: Practice ideas for tonight

- Add a search_policy tool and a prompt template to the MCP server.
- Add a second skill to the A2A refund agent and route by skill name.
- Add a timeout and a row limit to the data analysis sandbox.

Coming up on Day 28: LLM security and guardrails: prompt injection, jailbreaks, output filtering, responsible AI and human-in-the-loop.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today